

Stage 16N Report

Updated 2026-06-21

1 Overview

Stage 16N investigates restart-preserved cycle-jump behavior for the Chaboche UMAT Abaqus benchmark. The R4H diagnostic tested whether source-split restart files differ from restart records produced by long restarted replay runs. The R4I-R recovery rerun then repeated the cycle-280 source-split branch with durable lightweight copy-back and absolute reference CSVs.

2 Results

R4H completed six PBS/Abaqus jobs successfully with `Exit_status=0`. Long-replay restart records passed exactly: R4H1 at cycle 500, R4H3 at cycle 500, and R4H5 at cycle 750 all had zero global, local, and diagnostic S11 error. Short source-split restart cases remained inconsistent: R4H2 failed at cycle 500 with 11.83033% max primary local error, R4H4 was review-level at cycle 500 with 9.3860617%, and R4H6 was review-level at cycle 750 with 8.0369449%.

The R4I-R first wave finished by 2026-06-20 13:28 CEST and was absent from the live PBS queue on 2026-06-21 05:47 CEST. PBS history no longer returned records for jobs 1350601 and 1350602, so the current classification uses scratch case status files, PBS stdout, Abaqus completion messages, and copied comparison CSVs. R4I-R1 passed exactly at cycle 500; R4I-R2 failed with the same source-split signature as R4H2.

Case	Mode	Endpoint	Status	Global %	Primary local %	S11 %
R4H1	Long replay 280–500	500	pass	0	0	0
R4H2	Source 250–281, restart 280–500	500	fail	1.1887403	11.83033	0.92936729
R4H3	Long replay 270–500	500	pass	0	0	0
R4H4	Source 250–271, restart 270–500	500	review	2.5314199	9.3860617	1.2392968
R4H5	Long replay 505–750	750	pass	0	0	0
R4H6	Source 500–506, restart 505–750	750	review	0.31896796	8.0369449	1.5702038

R4I-R first-wave recovery

Case	Source solve	Restart	Endpoint	Status	Global %	Primary local %	S11 %
R4I-R1	Deck-clone 250–281	280–500	500	pass	0	0	0
R4I-R2	Buffered 250–300	280–500	500	fail	1.1887403	11.83033	0.92936729

R4I-R1 shows that cloning/truncating the clean direct replay deck shape repairs the cycle-280 branch. R4I-R2 shows that a longer buffered generated source alone does not repair that branch; the dominant mismatch remains `HOLE_RING_SDV8_MAX`.

R4I-R second wave and deck-clone confirmations

After the first-wave classification, R4I-R3 and R4I-R4 were submitted on 2026-06-21 through the guarded storage-gated scratch workflow. Both jobs request one node with 16 cores, one MPI rank, 16 OpenMP threads, 90 GB memory, and 24 h walltime. R4I-R3 tests the generated buffered source at restart 270; R4I-R4 tests the generated buffered source at restart 505.

Two deck-clone confirmation cases were prepared but not submitted. R4I-R5 clones/truncates the clean replay source from 250–271 and restarts at cycle 270 for continuation to 500. R4I-R6 clones/truncates the clean replay source from 500–506 and restarts at cycle 505 for continuation to 750. These cases should run only after R4I-R3/R4I-R4 finish or no longer consume the two active Stage 16N job slots.

3 Images

No R4H images were generated in this update.

4 Tables

The key R4H and R4I-R comparison tables are included in the Results section. Lightweight R4I-R first-wave evidence and the prepared R4I-R5/R4I-R6 decks are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/  
stage16n_restart_control/r4ir_restart_source_buffer_recovery/
```

The earlier R4H CSV evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/  
stage16n_restart_control/r4h_restart_source_diagnostics/
```

5 Failures and Debugging

No R4H job failed at PBS/Abaqus level. The original R4I comparison failed because remote reference CSV paths were missing after extraction, and its scratch outputs were not recoverable. R4I-R fixed that workflow by staging absolute reference CSVs and copying lightweight evidence before comparison. The scientific failure now narrows to generated source-split consistency: R4I-R2 reproduces the R4H2 error even with a larger 250–300 source buffer.

6 Conclusion and Next Steps

R4H shows that long-replay restart records are clean, while short source-split restart generation or source-split phase history is still inconsistent. R4I-R1 proves that a deck-clone source can pass exactly at cycle 500, while R4I-R2 proves that a larger buffered generated source is still insufficient at restart 280. R4I-R3/R4I-R4 are now the active second-wave generated-buffer tests. Do not submit R4J9/R4J10 yet. After R4I-R3/R4I-R4 are classified, run R4I-R5/R4I-R6 as deck-clone confirmations; if those pass, rerun exact controls with the deck-clone source workflow before any new extrapolated true jump.